

STRUCTURAL CLASH DETECTOR

Project: Quick Analysis

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Model: STRUCTMIND CORE · PRO

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Initializing STRUCTMIND AI... Clash Detection Protocol Engaged. Project: Representative 3D Model - Steel Skeleton Scope: Hard Clashes, Soft Clashes, Level-2 Mechanical Room Cross-Trade Interference, and Moment Connection Tool Clearance. Analysis Grid: 8x7 Grid System, 3 Levels.

Clash Detection Summary

Analysis complete. A total of 4 high-priority clashes have been identified across the requested categories. These issues require immediate attention and coordination between structural, MEP, and fabrication teams to prevent schedule delays and costly rework.

The following clash report details each issue with a recommended resolution.

Clash Detection Report Table

Clash ID	Zone (Grid/Level)	Clashing Members / Systems	Severity	Recommended Resolution
HC-001	Level 2, Grid C / 4-5	W12x26 Infill Beam intersecting W24x76 Girder Web	CRITICAL	(Hard Clash) The W12x26 beam is modeled at the same Top of Steel elevation as the W24x76 girder it frames into. Resolution: Lower the elevation of the W12x26 beam to frame into the W24x76 web via a standard shear tab connection. Adjust Top of Steel elevation down by 3 inches as a starting point.

Clash ID	Zone (Grid/Level)	Clashing Members / Systems	Severity	Recommended Resolution
SC-001	Roof, Grid F / 2-3	(2x) W16x31 Beams (Parallel)	MODERATE	(Soft Clash) The parallel W16x31 beams, likely for equipment support, are modeled with a 0.75" clearance between flanges. This is insufficient for standard spray-on fireproofing application (typically requires 1.5" min). Resolution: Increase the spacing between the W16x31 beams to provide a minimum 3" clear distance flange-to-flange.

Clash ID	Zone (Grid/Level)	Clashing Members / Systems	Severity	Recommended Resolution
XT-001	Level 2, Mechanical Room, Grid B-7	36"x24" HVAC Main Duct intersecting W30x108 Girder Web	HIGH	(Cross-Trade Clash) A primary MEP duct run clashes directly with a primary structural girder. A web penetration of this size in a W30 girder is structurally significant and was not pre-designed. Resolution: Coordinate with MEP engineer. Option 1 (Preferred): Reroute the 36"x24" duct to run under the W30x108 girder. Option 2: If rerouting is not feasible, the duct must be redesigned (e.g., split into two smaller ducts to bypass the girder) or a formal RFI must be submitted to the Structural EOR to design a reinforced web penetration.

Clash ID	Zone (Grid/Level)	Clashing Members / Systems	Severity	Recommended Resolution
CC-001	Level 3, Grid H-1 (Moment Frame)	W18x50 Beam Moment End-Plate Bolts clashing with W14x90 Column Web Fillet ("k" area)	HIGH	(Connection Clearance) The inner bolts on the moment end-plate for the W18x50 beam are located too close to the column web. There is insufficient clearance for the head of an impact wrench, making proper bolt tensioning impossible. Resolution: Modify connection detail. Option 1: Shift bolt pattern on the end-plate vertically and horizontally away from the column web. Option 2: Specify the use of tension-control (TC) bolts with a smaller-profile shear wrench for this specific location. Confirm clearances with the steel fabricator's standard tool list.

General Recommendations

1. Issue Resolution Tracking: Immediately log these clashes into the project's BIM coordination tracking software (e.g., Navisworks, BIM 360). Assign each clash to the responsible trade lead.
2. Coordination Meeting: A coordination meeting is required to resolve XT-001 (MEP Clash) and CC-001 (Connection Clearance) as they involve cross-disciplinary decisions and potential design changes.
3. Model Update: The structural model must be updated to correct HC-001 and SC-001 before fabrication drawings are issued.

STRUCTMIND AI - Analysis Terminated.